II- Assay questions.

II-1) In a follow-up to Beadle and Tatum's original study, they decide to identify the genes required for synthesis of another amino acid, Leucine (L). Like Arginine, Leucine is synthesized via a series of steps. Using X-rays you create Leucine auxotrophs. You identify 20 mutants. Complementation tests reveal 4 complementation groups: 1, 2, 3, 4. They suspect that the following molecules may be precursors to Leucine (L): isopropylmalate (IP), ketoisocaporate (KC), dihidroxyvalerate (DV), ketoisovalerate (KV). Each mutant is tested for growth on minimal media (mm) + each Leucine precursor. The results are below:

mutant	mm+ KC	mm+ IP	mm+KV	mm+ DV	mm+Leucine
1	no	grows	no	no	grows
2	no	no	no	no	grows
3	grows	grows	no	grows	grows
4	grows	grows	no	no	grows

Draw out the pathway indicating with arrows the order of each precursor and showing where each gene functions in the pathway. (4pts)

	KV	DV	KC	IP	Leu
1	−	−	−	+	+
2	−	−	−	−	+
3	−	+	+	+	+
4	−	−	+	+	+

KV–3--DV--4--KC--1--IP---2--Leu

II-2) The genetic analysis shown below describe the genotypic ratios of three genes in corn: an ("Anther ear"), br ("brachytic") and f ("fine stripe"). When a plant that is heterozygous for each of these markers is test-crossed with a homozygous recessive plant, the following results are obtained:

Test Cross	Progeny Numbers	
wild type	98	$an^+ \; br \; f$
fine	26	$an \; br^+ \; f^+$
brachytic	2	$an^+ \; br^+ \; f^+$
brachytic, fine	379	$an \; br \; f$
anther	385	$an^+ \; br^+ \; f$
anther, fine	3	$an \; br \; f^+$
anther, brachytic	20	$an \; br^+ \; f$
anther, brachytic, fine	87	$an^+ \; br \; f^+$ → middle.
Total Offspring:	1000	

1) Determine the map distance between the three genes. (3 pts)
2) Determine the order of these genes. (1pt)

$$an\text{-}br = \frac{98+87+26+20+(3\times2)+(2\times2)}{1000} = 0.241 \times 100 = 24.1 \text{ mu}$$

$$an\text{-}f = \frac{98+87+3+2}{1000} = 0.19 \times 100 = 19 \text{ m.u}$$

$$br\text{-}f = \frac{26+30+3+2}{1000} = 0.061 = 6.1 \text{ m.u}$$

an ———— f ———— br

II-2) **The sequence of a segment of mRNA, beginning with the initiation codon, is given here, along with the corresponding sequences from several mutant strains.**

Met Thr His ~~Arg~~ Gly Val Val Asn Pro Lys

Normal AUG ACA CAU CGA GGG GUG GUA AAC CCU AAG...

Mutant 1 AUG ACA CAU CCA GGG GUG GUA AAC CCU AAG...

Pro

Mutant 2 AUG ACA CAU CGA GGG UGG UAA ACC CU AAG...

Trp Stop

Mutant 3 AUG ACG CAU CGA GGG GUG GUA AAC CCU AAG...

Thr ~~Arg~~

Mutant 4 AUG ACA CAU CGA GGG GUU GGU AAA CCC UAA G...

Val Gly Lys Pro Stop

Mutant 5 AUG ACA CAU UGA GGG GUG GUA AAC CCU AAG...

Stop

First letter	U	C	A	G	Third letter
	Second letter				
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

***Indicate the type of mutation present in each, and translate the mutated portion of the sequence into an amino acid sequence in each case. Use the table to determine the amino acid of each codon (5pts)**

Mutant 1 :- ~~Silent mutation~~ Missense mutation.

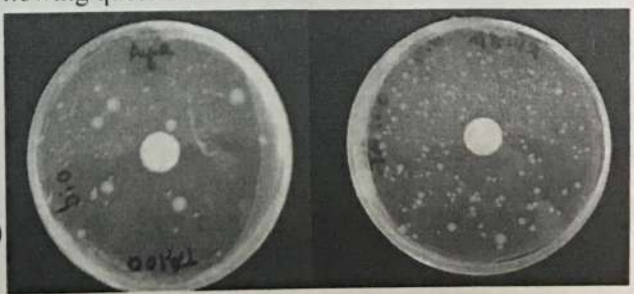
Mutant 2 : ~~Non Sense Mutation~~. deletion

Mutant 3 : ~~Missense~~ Silent Mutation.

Mutant 4 : Frameshift Mutation.

Mutant 5 :- Non sense Mutation.

II-3) Study the results of the Ames test below and answer the following question



Control (no chemical) Test (with chemical)

1) The bacteria used for this test is known as His⁻. What does this mean? (1pt)

auxotrophic bacteria can grow without histidine.

2) What can you conclude about the chemical tested?(1pt)

Bacteria doesn't produce histidine so it won't grow in media which lack histidine but when we add histidine to the media bacteria will grow.

3) Why do you see some bacterial colonies grown in the control although no histidine was added? (1pt)

Bacteria doesn't have histidine but the mutation that happened make bacteria can grow without Histidine.